

ELL715 - Assignment 4

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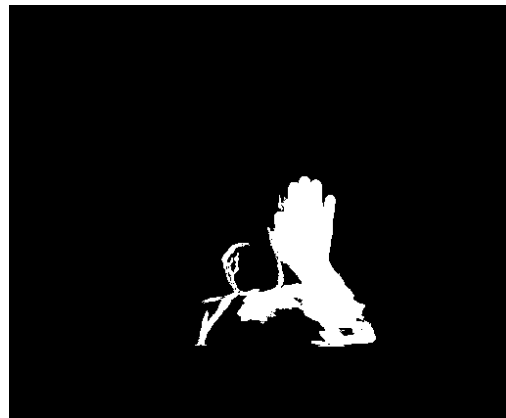
Date: October 18, 2025

1 Part1: Paper Implementation

1.1 Implementation of Silhouette Tunnel



(a) original-Compass

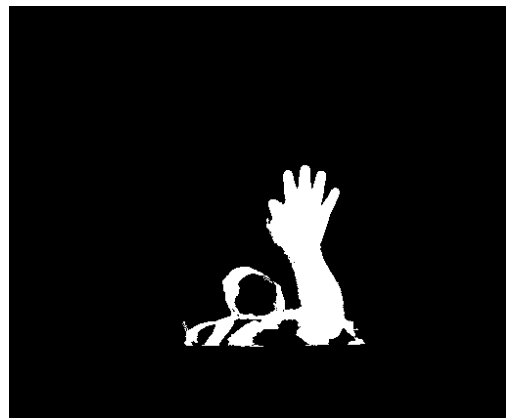


(b) silhouette mask - Compass

Figure 1: Extraction of silhouette mask from compass



(a) original-Piano



(b) silhouette mask - Piano

Figure 2: Extraction of silhouette mask from piano



(a) original-Push

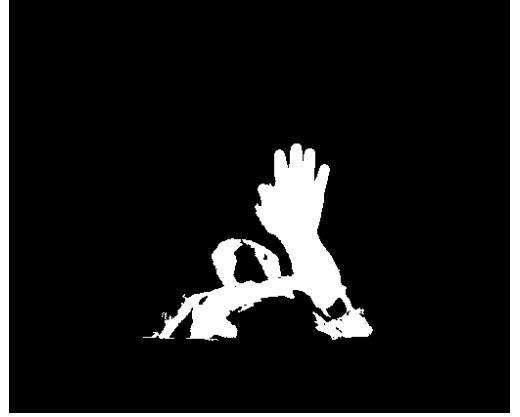


(b) silhouette mask - Push

Figure 3: Extraction of silhouette mask from Push



(a) original-UCDO



(b) silhouette mask - UCDO

Figure 4: Extraction of silhouette mask from UCDO

Each pixel within the silhouette tunnel is represented by a 14-dimensional feature vector:

$$\mathbf{f} = [x, y, t, z, d_E, d_W, d_N, d_S, d_{NE}, d_{NW}, d_{SE}, d_{SW}, dT^+, dT^-]^T$$

where:

- x, y, t denote the spatial and temporal coordinates of the pixel,
- z is the depth value,
- $d_E, d_W, d_N, d_S, d_{NE}, d_{NW}, d_{SE}, d_{SW}$ represent directional distances from the pixel to the nearest background boundary in the eight cardinal directions, and
- dT^+, dT^- denote the forward and backward temporal persistence of the silhouette pixel across time.

Directional distances and temporal runs are efficiently computed using Numba-accelerated kernels for parallel processing.

All feature vectors from the sequence are collected into a feature matrix:

$$\mathbf{F} = [\mathbf{f}_1 \quad \mathbf{f}_2 \quad \dots \quad \mathbf{f}_N] \in \mathbb{R}^{14 \times N}$$

and are normalized feature-wise to the range $[0, 1]$ as:

$$\mathbf{F}_{\text{norm}} = \frac{\mathbf{F} - \mathbf{F}_{\min}}{\mathbf{F}_{\max} - \mathbf{F}_{\min} + \epsilon}$$

where ϵ is a small constant (e.g., 10^{-6}) to avoid division by zero.

1.2 Extraction of Covariance Matrix

The silhouette tunnel provides us with a rich representation of the pixel features over time. To analyze the variability and relationships between these features, we compute the covariance matrix of the normalized feature matrix:

$$\mathbf{C} = \frac{1}{N-1} \mathbf{F}_{\text{norm}} \mathbf{F}_{\text{norm}}^T$$

where $\mathbf{C} \in \mathbb{R}^{14 \times 14}$ is the covariance matrix capturing the pairwise feature correlations across all pixels in the silhouette tunnel across all the frames.

Note: In the paper, the denominator of the above equation is given as N . However, to obtain an unbiased estimate of the covariance, we use $N-1$ in our implementation.

We then just take the upper triangular part of the covariance matrix (including the diagonal) and vectorize it to form our final feature descriptor for the video sequence:

$$\mathbf{d} = \text{vec}(\mathbf{C}_{\text{upper}})$$

where $\mathbf{d} \in \mathbb{R}^{105}$ is the final feature descriptor capturing the relationships between the pixel features over time.

1.3 Temporal Hierarchical Covariance Descriptor

To capture the temporal dynamics of each gesture, the feature matrix $F \in \mathbb{R}^{14 \times N}$ is divided into multiple temporal segments. We use a three-level hierarchy where the sequence is split into $1 + 2 + 4 = 7$ parts in total, giving both coarse and fine temporal resolution.

For each segment, a covariance matrix is computed as:

$$C = \frac{1}{N-1} \sum_{n=1}^N (f_n - \mu)(f_n - \mu)^\top, \quad \text{where} \quad \mu = \frac{1}{N} \sum_{n=1}^N f_n$$

Each C is symmetric, so we only keep its upper-triangular entries (105 values for 14 features). The seven such vectors are then concatenated to form the final descriptor:

$$\text{Descriptor}_{TH} = [C_1^{ut}, C_2^{ut}, \dots, C_7^{ut}] \in \mathbb{R}^{735}$$

This representation captures how feature correlations evolve over time while remaining compact.
Implementation Note: all covariance computations are parallelized using Numba for efficiency.

Result Note: There has been barely an improvement in the results after implementing temporal hierarchy. The table showing the results before and after implementing temporal hierarchy is as follows:

Gesture	Without Temporal Hierarchy	With Temporal Hierarchy
Compass	0.280	0.289
Piano	0.305	0.303
Push	0.280	0.275
UCDO	0.298	0.275

Table 1: Comparison of Recognition Accuracy With and Without Temporal Hierarchy using EER

1.4 Hand-Morphology:

For each sub-silhouette, a separate feature matrix $F_k \in \mathbb{R}^{14 \times N_k}$ is computed, following the same 14-dimensional feature representation used for the full silhouette. Each sub-silhouette then produces its own temporal-hierarchical covariance descriptor $C_k \in \mathbb{R}^{735}$.

The final multichannel descriptor is formed by concatenating the descriptor of the full silhouette with those of the $K = 3$ sub-silhouettes:

$$\text{Descriptor}_{multi} = [C_{full}, C_1, C_2, C_3]$$

This results in a combined representation of dimension:

$$4 \times 735 = 2940$$

Some examples of subsilhouettes representation:

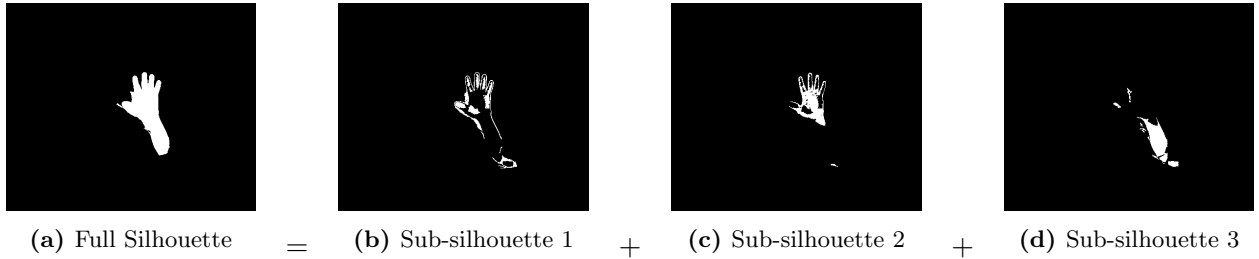


Figure 5: Example of sub-silhouette decomposition from a full silhouette for the compass gesture.

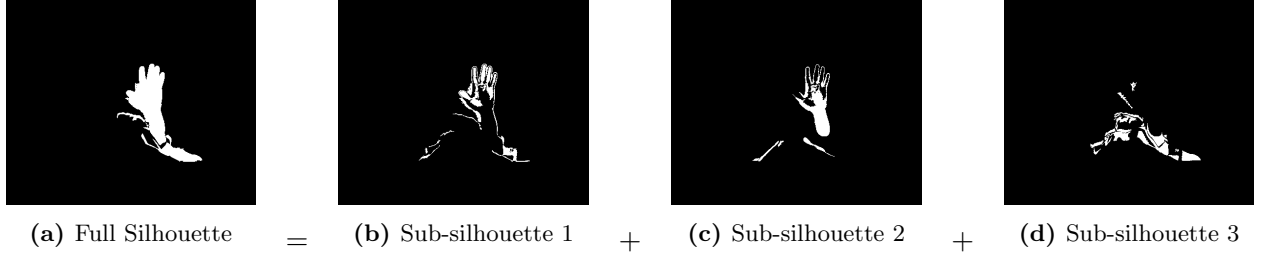


Figure 6: Example of sub-silhouette decomposition from a full silhouette for the piano gesture.

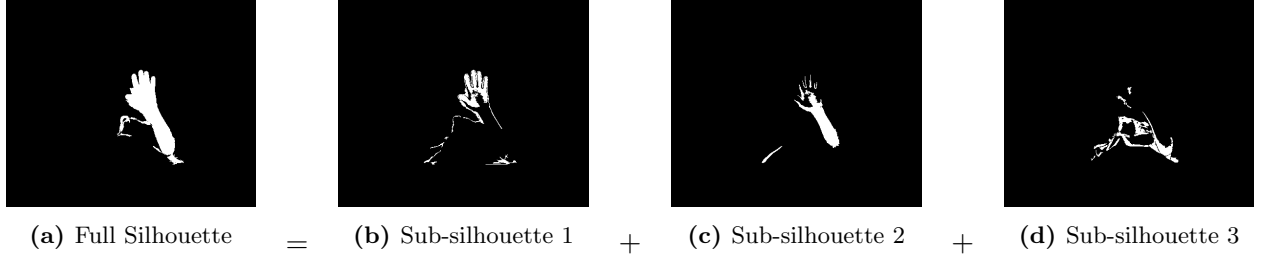


Figure 7: Example of sub-silhouette decomposition from a full silhouette for the push gesture.

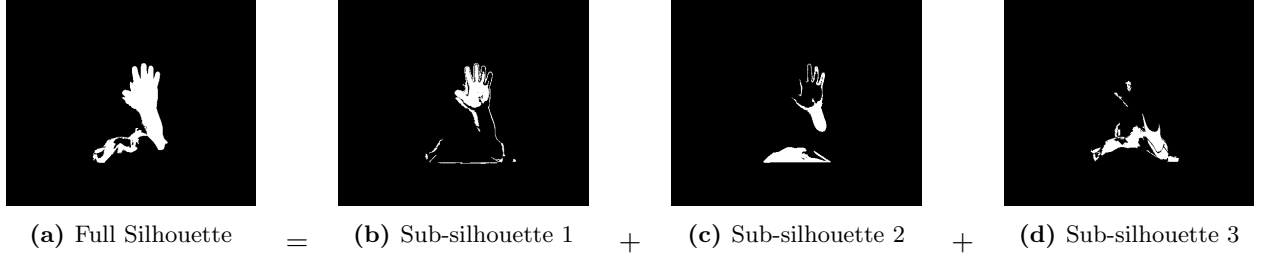


Figure 8: Example of sub-silhouette decomposition from a full silhouette for the UCDO gesture.

Note: The above study has been done for test sample 001 of user 02, 2nd frame, considering the 1st frame as background; for all the gestures, i.e., Compass, Piano, Push and UCDO.

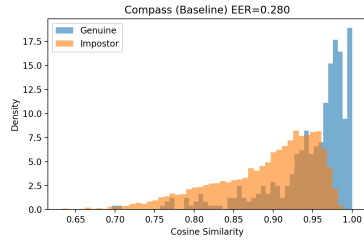
For cosine similarity:

$$d_{\text{cosine}} = 1 - \frac{1}{4} \sum_{k \in \{\text{full}, 1, 2, 3\}} \frac{\langle C_k^{(a)}, C_k^{(b)} \rangle}{\|C_k^{(a)}\|_2 \|C_k^{(b)}\|_2}$$

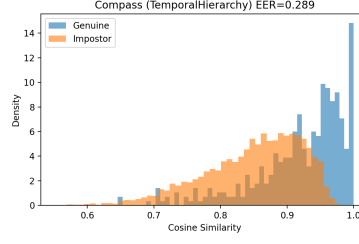
where $C_k^{(a)}$ and $C_k^{(b)}$ are the covariance descriptors of the k -th channel for samples a and b .

1.5 ablation study

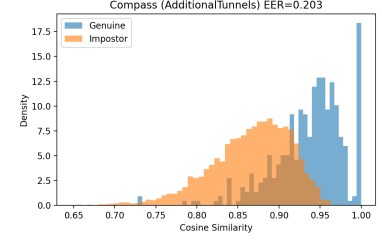
We have considered three scenerios for the ablation study, which includes; Baseline, Baseline + Temporal Hierarchy, and Baseline + Temporal Hierarchy + Hand Morphology. The results are as follows:



(a) Baseline

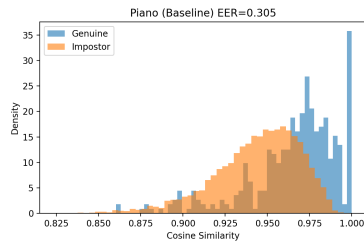


(b) Baseline + Temporal Hierarchy

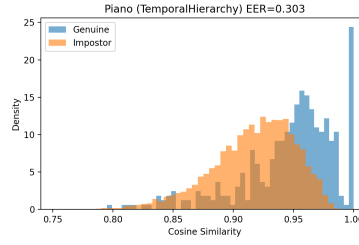


(c) Baseline + Temporal Hierarchy + Hand Morphology

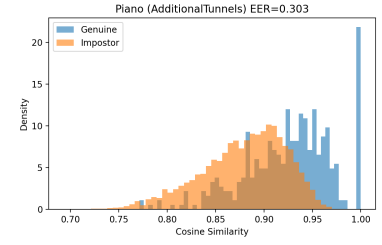
Figure 9: Ablation Study Confusion Matrices for Different Scenarios



(a) Baseline

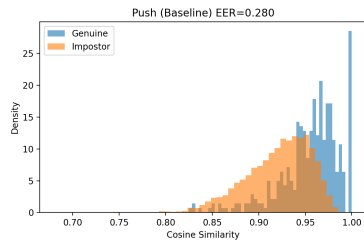


(b) Baseline + Temporal Hierarchy

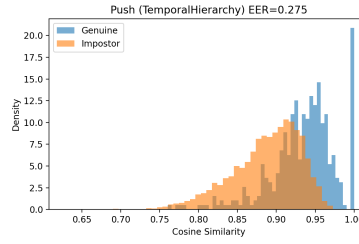


(c) Baseline + Temporal Hierarchy + Hand Morphology

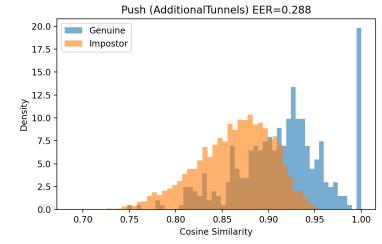
Figure 10: Ablation Study Confusion Matrices for Different Scenarios - Piano Gesture



(a) Baseline

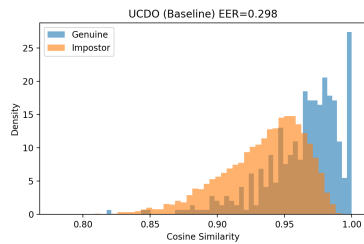


(b) Baseline + Temporal Hierarchy

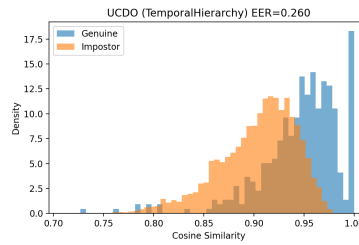


(c) Baseline + Temporal Hierarchy + Hand Morphology

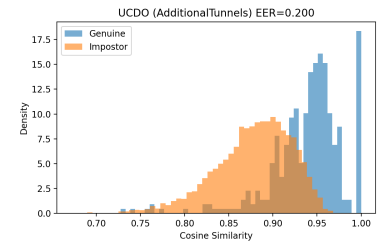
Figure 11: Ablation Study Confusion Matrices for Different Scenarios - Push Gesture



(a) Baseline



(b) Baseline + Temporal Hierarchy



(c) Baseline + Temporal Hierarchy + Hand Morphology

Figure 12: Ablation Study Confusion Matrices for Different Scenarios - UCDO Gesture

Table 2: Equal Error Rate (EER) comparison across different ablation settings and gestures.

Gesture	Baseline	Temporal Hierarchy	Hand Morphology
Compass	0.280	0.265	0.204
Piano	0.305	0.303	0.280
Push	0.260	0.275	0.200
UCDO	0.299	0.260	0.226

References